

Scalability & Future Plan

Date	01 July 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Static Machine Learning Model	The model predicts using the trained dataset and requires retraining when new data becomes available.	High
2	No Prediction History	Users cannot view or compare previous prediction results.	Medium
3	Limited Input Parameters	The system predicts HDI using only four development indicators, which may limit prediction flexibility.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports local prediction requests through the Flask web application.	Deploy on a cloud platform with auto-scaling to handle higher traffic.
2	Data Storage	Uses a CSV dataset and trained model files (.pkl).	Store datasets and prediction history in a relational database (MySQL/PostgreSQL).
3	Performance	Uses a pre-trained Linear Regression model for fast predictions.	Optimize model loading and cache frequently used resources to improve response time.

4	Security	Basic input validation is implemented in the application.	Add user authentication, HTTPS, and stronger input validation for enhanced security.
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Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Prediction History	1-2 Months	Users can view and compare previous HDI predictions.
Phase 3	Updated Dataset & Model Retraining	2-3 Months	Improves prediction accuracy using the latest development data.
Phase 4	Interactive Dashboard & Charts	4-6 Months	Provides better visualization and comparison of HDI indicators.